

W10 Exercises



In the exercises, we will be using the functions we learned in W5!

Task 1: Central Limit Theorem (CLT) Simulation

1.1 Draw Poisson Sample

- Write a function called `sample_mean_pois` with the following specification:
 - Draw `n` random samples from `Poisson( $\lambda = 5$ )`
 - Compute the sample mean
 - Return the sample mean

1.2 Build Sample Mean Dataframe

- Now, we're going to utilize the function to help us compute the sample mean for multiple times. Please follow the instructions step-by-step:
 - Set the random seed as `2026`
 - Given `n = 500`, use the function `replicate()` to execute the function `sample_mean_pois()` we just defined for `1000` times
 - Now you obtain `1000` sample means. Assign the sample means to a variable called `rdmean` and build a dataframe from it

1.3 Histogram

- Use the dataframe we just built, draw a histogram to examine the distribution of the sample means